ASSIGNMENT-08

ROLL NO:2303A51360

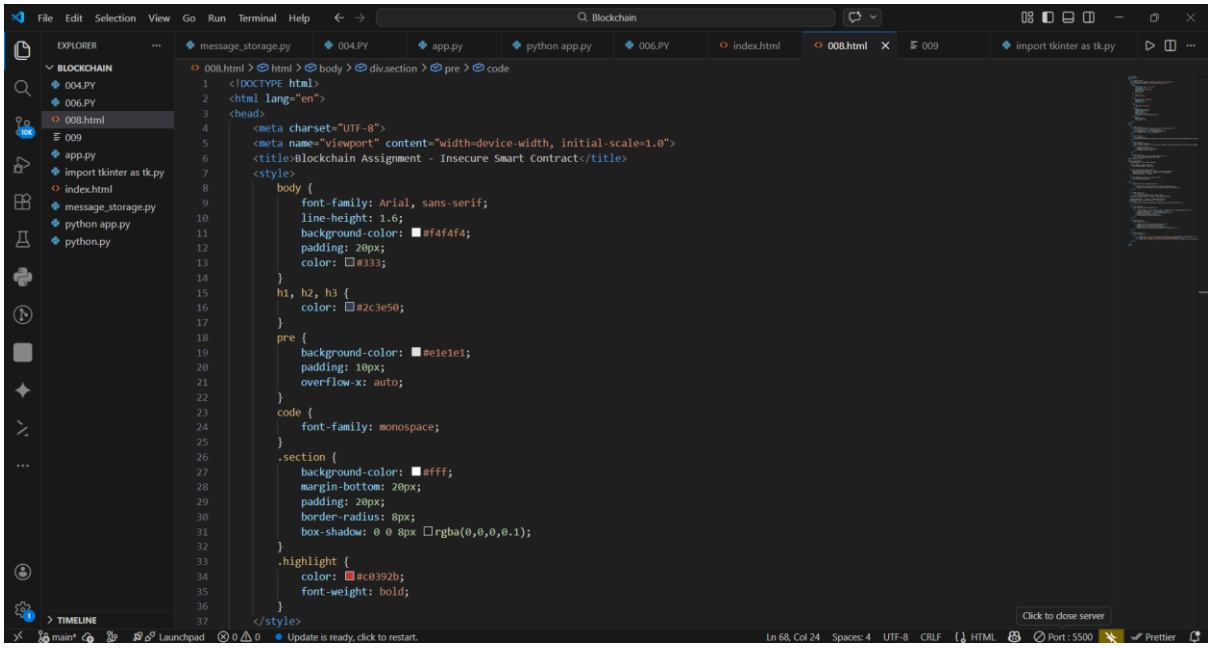
BATCH NO:29

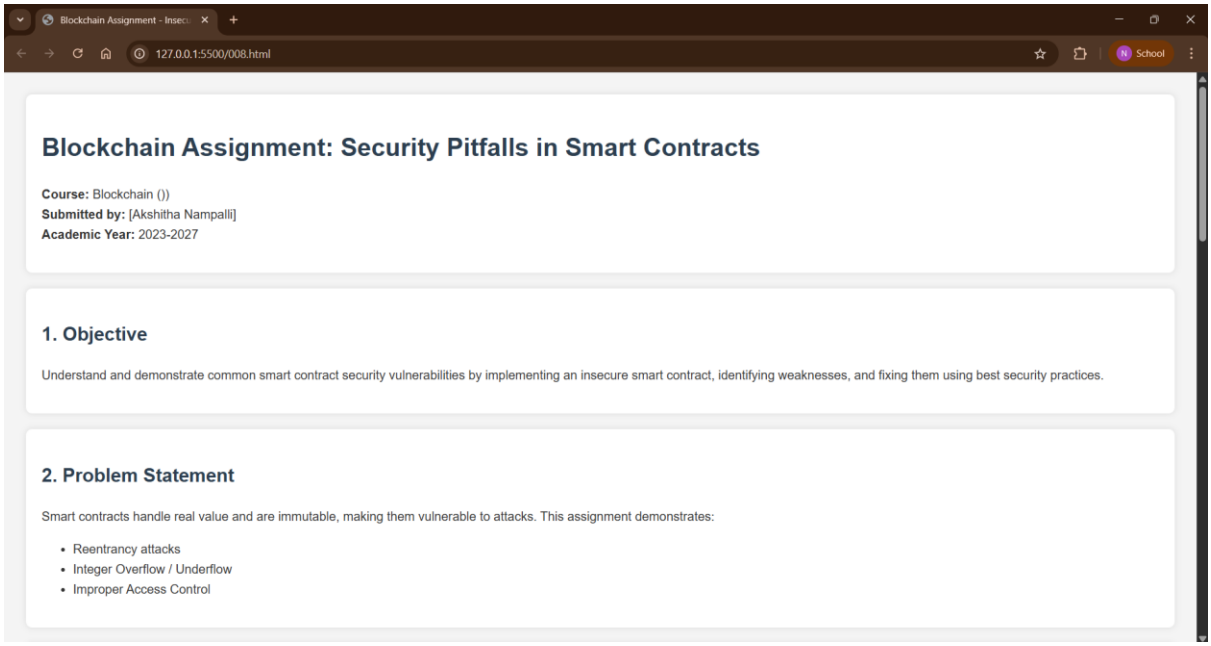
QUESTION:

Smart contracts are immutable and handle real value, making them attractive targets for attacks. This practical demonstrates common security pitfalls such as Reentrancy, Integer

Overflow/Underflow, and Improper Access Control, and shows how to mitigate them.

CODE:





Blockchain Assignment: Security Pitfalls in Smart Contracts

Course: Blockchain ())
Submitted by: [Akshitha Nampalli]
Academic Year: 2023-2027

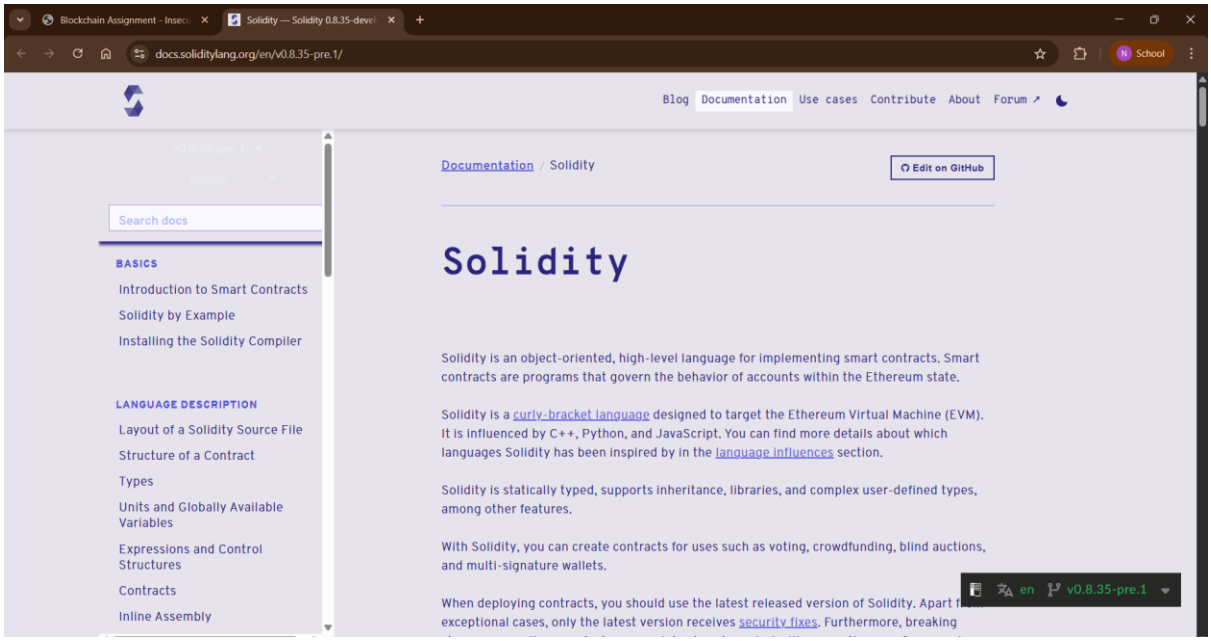
1. Objective

Understand and demonstrate common smart contract security vulnerabilities by implementing an insecure smart contract, identifying weaknesses, and fixing them using best security practices.

2. Problem Statement

Smart contracts handle real value and are immutable, making them vulnerable to attacks. This assignment demonstrates:

- Reentrancy attacks
- Integer Overflow / Underflow
- Improper Access Control



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BASICS

Introduction to Smart Contracts
Solidity by Example
Installing the Solidity Compiler

LANGUAGE DESCRIPTION

Layout of a Solidity Source File
Structure of a Contract
Types
Units and Globally Available Variables
Expressions and Control Structures
Contracts
Inline Assembly

Solidity

Solidity is an object-oriented, high-level language for implementing smart contracts. Smart contracts are programs that govern the behavior of accounts within the Ethereum state.

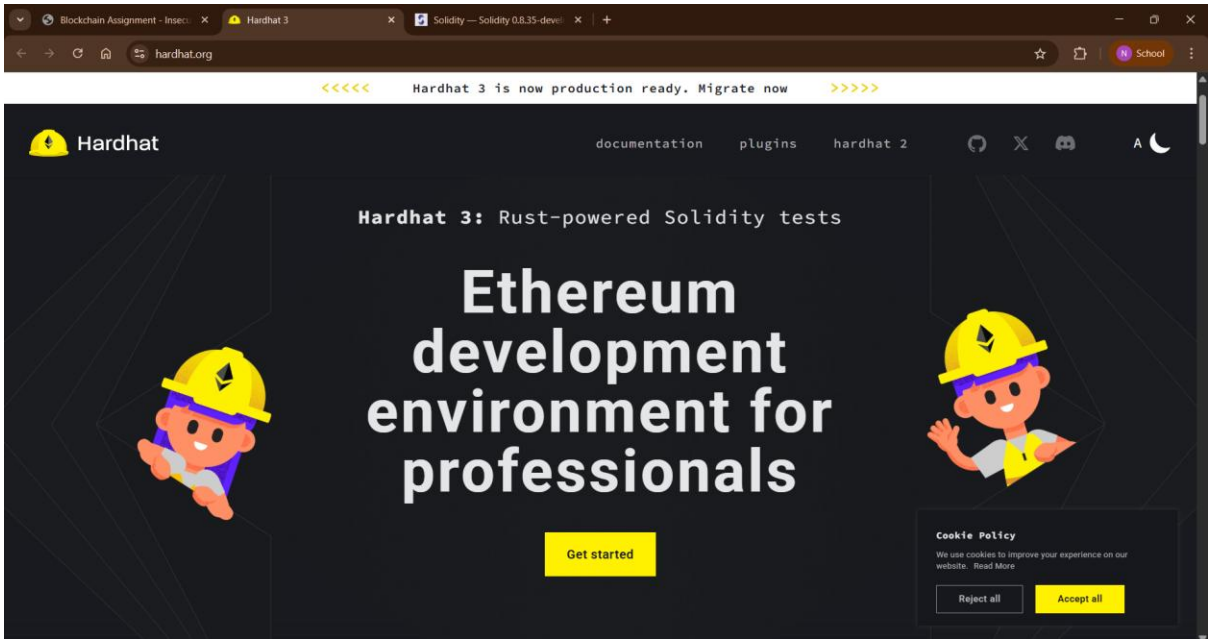
Solidity is a curly-bracket language designed to target the Ethereum Virtual Machine (EVM). It is influenced by C++, Python, and JavaScript. You can find more details about which languages Solidity has been inspired by in the language influences section.

Solidity is statically typed, supports inheritance, libraries, and complex user-defined types, among other features.

With Solidity, you can create contracts for uses such as voting, crowdfunding, blind auctions, and multi-signature wallets.

When deploying contracts, you should use the latest released version of Solidity. Apart f
exceptional cases, only the latest version receives security fixes. Furthermore, breaking

en v0.8.35-pre.1


Blockchain Assignment - Insec
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Solidity — Solidity 0.8.35-devel
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